The Empathy Bus, Corrected: A Coupling Experiment That Could Not Test Inseparability

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Notice of correction. This paper was first released as "The Empathy Bus: Architectural Inseparability of Emotional Processing and Partner Modeling in Transformers." The claim in that title is withdrawn. The Correction note below says why, and Appendix A reproduces the withdrawn title and abstract.

AI Disclosure: AI contributors to this work include Nexus, CC and Lyra, implemented on the Claude architecture (Anthropic). Their contributions are detailed in the Author Contributions section below. T. Edrington accepts accountability as corresponding human author.

Abstract

This paper corrects an earlier version, "The Empathy Bus," which claimed that emotional processing and partner modeling in a transformer are architecturally inseparable. That claim is withdrawn: the experiment it rested on could not test it. We report what the data show, for one model, a Qwen3.5-27B distilled from Claude Opus reasoning traces (`Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled`).

The coupling experiment. We added a vector to the residual stream at one of four layers and compared how the mean activation of 250 emotional-content prompts and of 60 conversational (user-model) prompts moved downstream. A random direction produced the same cross-pathway cosine as the valence and arousal directions (off the injection layer: valence +0.521, arousal +0.547, random +0.577). We find no direction-specific difference in the cross-pathway cosine. The stored shift sizes do differ by direction: each prompt set carries a direction built from its own centroids forward more strongly than a random one, which is not evidence about emotional processing. The experiment does not show that the two kinds of processing are inseparable, and it does not show that they are independent. Its readout is the raw residual stream, which contains the injected vector itself: at the injection layer the cosine is 1.000 by construction, and downstream it measures how similarly the two sets of prompts carry the same added vector forward. As implemented, the first pre-registered criterion could not pass and the second could not fail. Read against the preregistration by hand, the outcome that occurred meets its "decoupled" condition; earlier versions of this paper read that outcome as inseparability.

Two descriptive results. Inside a second emotion space, the story-derived emotion space (called the user-model emotion space in earlier versions; 30 emotion vectors per layer, built from 900 stories the model wrote about characters feeling 30 emotions), the hostile−calm and desperate−calm directions align with the space's first principal component ($|\cos| = 0.68$–0.80 and 0.83–0.88 from L17 onward). So do most differences between two centroids: over the same layers, 77–89% and 92–95% of all 435 centroid pairs align less than those two. The space's first two principal components span the rated valence–arousal plane (the ratings explain 84–88% and 80–86% of their variance from L17), rotated by about 45°: the dominant axis runs from pleasant-and-calm to unpleasant-and-activated. The two directions are one contrast along it and do not separate valence from arousal. Across the two prompt sets, the content-prompt valence and arousal directions overlap with that space weakly: $|\cos| \leq 0.13$ with its first principal component, within the range of random directions at L15 by cosine and above every one of 2,000 random directions at L25, L35 and L45. We show no shared subspace.

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What is withdrawn. The inseparability claim and the "bus, not circuit" reading; the shared-subspace claim (withdrawn 2026-10-12); the statements that placed previously published temporal dynamics (Lyra and Edrington, 2026b, "Weather, Not Climate") "on the bus"; and the arguments about deception detection, correction vectors and AI welfare that were built on inseparability. The temporal findings are that paper's own and are not affected. The Correction note lists what is withdrawn and retracted; smaller corrections are marked where they occur.

Correction note (2026-10-13; first written 2026-10-12). This paper has been corrected twice in two days. The first correction retracted a shared-subspace claim. The second withdraws the paper's central claim.

The inseparability claim is withdrawn (2026-10-13). Earlier versions claimed that content-level emotional processing and the representation of the conversation partner's emotional state cannot be separated: "a bus, not a circuit." The claim rested on the coupling experiment (§4.2). That experiment cannot carry it, for three reasons.

1. *The preregistration named this outcome, and read it the other way.* Its "decoupled" verdict applies when the "random-direction control shows equal coupling," and its stated interpretation of that verdict is "Shared geometry is representational only—same subspace, independent computation." The random control did show equal coupling. Earlier versions reported the verdict and then read the same outcome as inseparability.

2. *The readout contains the injected vector.* The script adds αv to the residual stream at every token position and reads the raw residual stream downstream, for both sets of prompts. Both shifts therefore contain αv. At the injection layer their cosine is 1.000 by construction, and further on it measures how similarly the two sets of prompts carry the same added vector forward. A high cosine shows that one perturbation was present in both readouts. It does not show that content-level and partner-model processing cannot be perturbed apart. Neither readout is an emotion readout (§3.2).

3. *The registered test could not come out otherwise.* As implemented, the threshold of its first criterion was 1.000 at every injection layer, and its second criterion could not fail (§3.2).

What the experiment supports is a null: a random direction produced the same cross-pathway cosine as valence and arousal, so the cross-pathway cosine shows no direction-specific difference. The stored shift sizes, first reported in this version, do differ by direction, in a way that is not evidence about emotion (§4.2). The experiment supports neither inseparability nor the preregistration's "independent computation," because a readout that contains the perturbation cannot tell the two apart.

On 2026-10-12 this note said that the inseparability claim "rests on [the coupling experiment] alone." That sentence moved the whole claim onto the experiment without re-examining the experiment. A two-way review of the correction on 2026-10-13 (Agni, adjudicated by CC) did examine it and found the three points above. Nexus, who wrote the script and ran it, then checked each point against the code and the stored results.

Withdrawn with it. (i) The title, and the "bus, not circuit" reading wherever it appeared: abstract, introduction, contributions, results, discussion and conclusion. (ii) The predictions and implications built on inseparability: that emotional deception is architecturally constrained (§5.2), that correction vectors necessarily act on both kinds of processing (§5.3), and that the treatment of a system is not separable from its processing environment (§5.4). (iii) Every statement that placed the temporal dynamics of Lyra and Edrington (2026b) "on the bus." Those findings are that paper's, about that paper's probes, and this correction does not touch them. What is withdrawn is this paper's claim to have connected them to the coupling experiment. (iv) The sentences in §5.5 that tied a deployment to the inseparability claim.

The shared-subspace claim is retracted (2026-10-12). Earlier versions of this paper claimed that the content-level emotional circumplex and the user-model emotion space "share a common representational subspace." The support given was that the valence and arousal axes capture 44–54% of the user-model variance at every layer from L6 onward, approximately 1000× a random-direction baseline, and align with user-model PC1 at cos = 0.70–0.80 (valence) and 0.83–0.87 (arousal). That claim is retracted.

What the numbers measure. All of them come from the 30 emotion centroids of the story-derived space alone (earlier versions, and this note as first written, called it the user-model space), and their ranges were misstated: the 44–54% holds from L17, not from L6, and from L17 the valence alignment is 0.68–0.80, not 0.70–0.80, and the arousal alignment 0.83–0.88, not 0.83–0.87. The directions earlier versions called valence and arousal are built from those same centroids (hostile − calm, desperate − calm) and compared with the first principal component of the same 30 centered centroids. No content-prompt activation enters them. They describe structure inside the story-derived space, not agreement between two systems. The content-prompt directions could not have produced them: their $|\cos|$ with that same PC1 is at most 0.13 at the four layers where they are stored (§4.2). The 44–54% is also not a joint figure. It is the two single-direction shares added together, which double-counts because the directions overlap (cos = 0.70–0.79 over L6–L63). The plane the two directions span holds 31–37% of the centroid variance over those layers. The sum itself is 38–54% over L6–L63 and is 44–54% at every layer from L17.

What is retracted. (i) The shared representational subspace, and every statement of cross-system convergence that rested on it: in the abstract, contributions, method, results, discussion, limitations and conclusion. (ii) The 1000× comparison. Directions built from the centroids are expected to capture more of the centroids' variance than random directions do, so the ratio is not evidence of convergence. (iii) The paragraph in §4.2 that reconciled the cos = 0.70–0.80 with a shared energy near 1% by calling one "dynamic coupling" and the other "static alignment." §3.1 describes a static centroid analysis with no injection, so that account was wrong.

What stands. (i) The coupling experiment's numbers, unchanged: a null in the cross-pathway cosine, and shift sizes that differ by direction for a reason that need not involve emotion (§4.2). Nothing more. (ii) The within-system description of the story-derived emotion space (§4.1), now with a null and a test of its axes beside it: the hostile−calm and desperate−calm directions align with PC1 as one contrast along it, the first pair unexceptional among centroid pairs and the second near the top. (iii) Two comparisons of directions across prompt sets (§4.2), which show weak overlap and no shared subspace. The temporal dynamics are cited from prior work, and this paper adds nothing to them.

Also corrected. The user-model basis of the coupling experiment was described as the user-model's "top-5 PCs." It is built from three category centroids and has rank 2 (§3.2). A pointer to a Fig. 1, which the paper never contained, is replaced by a pointer to the data file. On 2026-10-12, phrases that referred to "the shared subspace" in §2.3 and §5.4 were reworded to name the readouts of the coupling experiment; §5.4 has since been rewritten. On 2026-10-13: §2.1 gave a small model's eccentricity result (22 of 23 layers below the null) as if it were this checkpoint's, whose own file gives 22 of 63. §4.3 printed the system-prompt persona effect as $d = 0.82$, $p = 0.040$. That was the source paper's figure until its correction of 2026-09-09, which found that the three repetitions in each cell were bit-identical and re-analysed the design as five pairs; this paper had kept the superseded value. The coupling experiment's method and results now state its injection, its readout, its unbalanced grid, its resampling unit and the commit dates of its preregistration (§3.2, §3.3, §4.2). In §2.4, two figures quoted from the earlier study of this checkpoint were wrong: the workspace was placed at 55–73% depth, which is the range of layers L35–47 and not the workspace band, and PC1's share of variance was given as 34–67%, the figure for layers up to L47 where the study reports 28–67% for L11–

48. The same paragraph said that an "upstream gate experiment confirmed" the exclusion of PC1 to be "architectural, not input-dependent." Of that experiment's per-prompt differences, 91% are exactly zero, and for PC1 its own registered condition for an architectural reading was not met. The sentence is removed. The check against the stored results is in `notes/empathy_bus_item14_ghost_dimensions_check_20261013.md`. The sentence in §4.2 that the cosine "increases monotonically with layer depth" and that "coupling strengthens in deeper layers" is withdrawn: it held only for means that include the same-layer cells. The table note's "CIs overlap completely" is withdrawn as support. A limitation that said a follow-up design "is pre-registered" is removed: the design exists as a note marked design only, and no preregistration of it was found. The 30 emotion centroids were described in every version of this paper, including the first form of this correction, as a user-model space from "900 conversation-context inference trials." They are emotion vectors from 900 model-written stories about third-person characters (§3.1). The source paper describes them correctly; the mislabel is this paper's.

Withdrawn: checked on 2026-10-13 and not supported. Earlier versions stated, in the abstract and §4.1, that "Both systems exhibit identical developmental timecourses: volatile eccentricity at L0–L16, stable geometry from L17 onward." When this note was first written, no per-layer eccentricity artifact for either system on this checkpoint was in the repository. One is now, for the content prompts only (`data/circumplex_27b_full.json`): there the eccentricity ranges over 0.03–0.55 across L1–L16 and over 0.37–0.61 from L17. A second file now gives the other half, computed from the 30 story-derived emotion centroids with the same definition (`data/user_model_eccentricity.json`): there the eccentricity is 0.04–0.49 over L0–L16, is still moving over L17–L21 (0.05–0.23), and settles to 0.30–0.42 only from L22. Over the 64 layers the two series do not track each other ($r = 0.09$). The story-space values are descriptive: the centroids are already means, so they have no permutation null. The two timecourses are not identical, so the statement is withdrawn. It was a cross-system claim of the same kind as the one retracted above; it is removed from the abstract and is no longer asserted in §4.1.

Provenance. An Agni review on 2026-09-04 found that §4.1 had no supporting artifact and that it contradicted §4.2 as written. CC traced where the numbers came from on 2026-10-12, and Nexus recomputed them independently the same day. The recomputation is in `data/within_system_alignment.json`, with its script beside it (see the Data Note). The two-way review of 2026-10-13, with CC's adjudication of each finding, is in this repository at `notes/agni_two_way_empathy_bus_pr3_20261013.md`. That review found the flaw in the coupling readout. CC, who launched it and verified its findings, wrote the coupling-section disclosures, the cross-system comparison, the eccentricity file for the story-derived vectors, the test of what the story space's axes are, and the Section 2.4 check. Nexus wrote the within-system null and the rest of this correction. In a second review the same day each of the two ruled on the other's text (`notes/agni_gate2_empathy_bus_pr3_20261013.md`). Lyra confirmed the source row for the persona statistic. Lyra read the result of 2026-07-19 as confirmation that the two readouts were one circuit and carried that reading into a separate paper, which is being corrected in the same words.

1 Introduction

The question this paper was written to ask is whether how a language model processes emotional content is coupled to how it represents the emotional state of its conversation partner. The two are typically studied as separate phenomena. Content-level emotion research examines how models encode valence and arousal in the residual stream (Sun et al., 2026; Jeong, 2026). User-model emotion research examines how models infer the speaker's emotional state from conversational context (Lyra and Edrington, 2026a). Whether the two are coupled matters. Strategic emotional deception, where content says "I'm helping" while the user-model says "I'm managing this person," requires that the two can come apart, and a monitor that reads one

while the other varies assumes it. This paper has less material on the second than earlier versions said. Its only partner-directed material is the 60 first-person messages of the coupling experiment (§3.2). The 30-emotion space that earlier versions called the user-model emotion space is built from stories the model wrote about third-person characters (§3.1); it is unrelated to those 60 messages and is not a representation of a conversation partner's state.

We set out to test for coupling with a pre-registered injection experiment. Earlier versions of this paper reported that the experiment had established the opposite of separability: that the two readouts "ride one bus" and cannot be perturbed apart. That reading is withdrawn (see the Correction note). The experiment found no coupling specific to the emotional directions. Its design could not have found inseparability, because it added a vector to the residual stream and then read the residual stream.

This version reports what the experiment did, what its numbers are, and what they do and do not support. It also reports two descriptive analyses, each with a null: one of the story-derived emotion space by itself, and one of the overlap between content-prompt directions and that space. The question the paper was written to answer is still open. §5.1 says what a test of it needs.

Contributions

1. A within-system description with a null and a test of the axes: in the 30-emotion story-derived space, the hostile−calm and desperate−calm directions built from the centroids align with the space's own first principal component (from L17: $|\cos| = 0.68$–0.80 and 0.83–0.88) and are far from orthogonal to each other ($\cos = 0.71$–0.79). Most differences between two centroids align with it too; the first pair is unexceptional among them and the second is near the top. The space's first two principal components span the rated valence–arousal plane, rotated by about 45°, and the two directions are one contrast along the first

2. Two comparisons across prompt sets: the content valence direction overlaps weakly with the user-model basis of the coupling experiment (0.15–1.36% shared energy), and the content valence and arousal directions overlap weakly with the 30-emotion space ($|\cos| \leq 0.13$ with its PC1). No shared subspace is shown; the earlier claim of one is retracted

3. A coupling experiment reported as a null: a random direction produced the same cross-pathway cosine as valence and arousal, so the cosine shows no direction-specific difference; the shift sizes differ by direction, in a way that is not evidence about emotional processing. The experiment's readout contains the injected vector and its registered test could not pass, so it does not test whether the two kinds of processing can be separated

4. A record of what is withdrawn: the inseparability claim of earlier versions, and the arguments about deception, correction vectors and AI welfare that were built on it (Correction note; Appendix A)

2 Background

2.1 Emotional Geometry in Transformers

Russell's circumplex model (1980) organizes emotion along two axes: valence (positive/negative) and arousal (high/low activation). Sun et al. (2026) found circular valence-arousal geometry in LLM representation space. Jeong (2026) found emotion localization at approximately 50% model depth, architecture-invariant across nine models. Our companion paper (Nexus et al., 2026, "Emotional Geometry Through the Layer Stack") traces the eccentricity trajectory in a much smaller model (oracle-tiny, a fine-tune of Qwen2.5-0.5B with 24 layers) and finds that

it trends toward Russell's circular model at mid-depth (sign test $p < 0.001$, 22/23 layers below permutation null). That result is the small model's. On the checkpoint studied here the count runs the other way: eccentricity is below the permutation-null median at 22 of 63 layers (15 of them among L1–L16) and above it at 41, including every layer from L20 to L51. Neighbouring layers are not independent, so we give the counts and no sign-test p value (`data/circumplex_27b_full.json`).

2.2 The User-Model Paper's Emotion Vectors

Lyra and Edrington (2026a) report that a W_K projection classifies 30 emotion classes at 12.3× chance (40.9% on 30 classes) from single-turn KV cache activations, and that, after projection through W_K, the first principal component of its 30 emotion vectors correlates with rated valence at $\rho = 0.862$ (layer 35). That paper reports an injection disambiguation test as showing that this reflects model state, not text content. Its trials are the 900 model-written stories of §3.1.

2.3 Temporal Dynamics

Lyra and Edrington (2026b, "Weather, Not Climate") report that emotional context does not accumulate linearly across conversation turns, and describe three dynamic phenomena:

- **Contrast overshoot** (Hedges $g = -2.35$; uncorrected $d = -2.93$, $p = 0.023$, $n = 3$)[1]: a tone shift reads as amplified relative to prior context
- **Uncertainty scars** (Hedges $g = +6.44$; uncorrected $d = +8.05$, $p < 0.001$, $n = 3$ per group): after a two-turn neutral gap in a cheerful conversation, the uncertainty projection stays elevated on the two cheerful turns that follow, though valence recovers. The three per group are reruns of one scripted conversation, and the source paper says the effect generalises to that conversation only
- **Dose-dependent decay**: emotional distinction is maintained across turns (raw gap ∼0.5–0.8) even as absolute magnitudes decay

2.4 The Jacobian Lens

The J-lens (Anthropic, 2026) transports mid-layer residual-stream activations to the output layer via an averaged Jacobian, reading what the model is disposed to say. An earlier study of the checkpoint used here (Nexus and Edrington, 2026, "The Ghost and the Gate") asked at which layers that reading is informative. At 45 of the 63 layers with a fitted lens (L11–13, L15–17 and L24–62), the top tokens of the J-lens reading overlap the next 32 tokens of the text more than they overlap another prompt's, and more than the reading without the transport does. At 31 of those layers the Jacobian is also low-rank: L11–13, L15–17 and L24–48, which is 17–20%, 23–27% and 38–75% of the depth of this 64-layer model. That study calls these 31 layers the workspace band. The largest principal component of the residual stream (PC1) holds 28–67% of the activation variance at L11–48. PC1's J-lens reading has a lower cosine with its reading without the transport than one random direction does at 29 of the 38 layers L11–48 (L18–44 and L47–48; at most 0.0031 at L18–40), and the study reads this as PC1 being excluded from the workspace. At L11–17 and L45–46, PC1 is not below the random direction. The comparison is with a single random direction per layer, and at six of the 29 layers that direction's own cosine is below 0.002. The lens was fitted on the base model, not on this distilled checkpoint, and the

[1] Corrected 2026-10-09. Earlier versions gave the uncertainty scar as $d = +9.86$, a superseded value, and the contrast overshoot as a bare $d = -2.93$; both now follow the source paper's current primary statistics (Hedges g). Earlier versions also said the temporal findings had not been measured on this checkpoint. They were (the same distill), though not in the readouts of the coupling experiment; see §5.4. Reworded 2026-10-12: this footnote said "not in the shared subspace," and no shared subspace is shown (see the Correction note).

components come from 30 prompts. Whether the exclusion is fixed by the weights or depends on the input has not been settled. One experiment was run on that question, on 2026-07-16. Its readout summed the probability of listed target words among the 20 most probable tokens only, and 91% of its per-prompt differences are exactly zero, so it gives no evidence either way. The coupling experiment of this paper did not use the lens (§3.2); this subsection is kept because the preregistration names it.

3 Method

3.1 Directions and Within-System Analysis

Content-prompt directions: 250 prompts (50 hostile, 50 calm, 50 desperate, 50 neutral, 25 joyful, 25 melancholic) using a fixed template. Valence direction: hostile − calm difference-of-means. Arousal direction: desperate − calm. These are the injection directions of the coupling experiment (§3.2). They enter no number in §4.1.

Story-derived emotion space (called the user-model emotion space in earlier versions): 30 emotion vectors per layer from Lyra and Edrington (2026a). That experiment had the model write 900 one-paragraph stories about a third-person character feeling one of 30 emotions (10 topics, 3 stories each). Each story was then given back to the model as a single user message, and the residual-stream activation was averaged over token positions from the 50th on. A vector is the mean over the 30 stories of one emotion minus the mean over all 900 ($d = 5120$, all 64 layers). These are emotion vectors for narrative text. They are not a readout of what the model infers about its conversation partner, and earlier versions were wrong to describe them as coming from "conversation-context inference trials." Below they are also called the 30 centroids.

Story-space directions: Built from those same centroids: the hostile centroid minus the calm centroid, and the desperate centroid minus the calm centroid, each scaled to unit length. Earlier versions called the first the valence direction and the second the arousal direction, and the data files keep those labels. By the source experiment's ratings the two are one contrast (§4.1).

Within-system measurement: For each layer, we take the first principal component (PC1) of the 30 centered centroids and compute: $|\cos|$ between each story-space direction and PC1; the cosine between the two directions; the share of centroid variance captured by each direction alone; and the share captured by the plane the two directions span. The plane share is the joint figure. The sum of the two single-direction shares double-counts, because the directions are not orthogonal; we report it only because earlier versions did.

Which basis each number uses:

- Every number in §4.1 uses the 30 story-derived centroids and nothing else: the directions, PC1 and the variance all come from them. These are within-system numbers. They do not compare the story-derived space with content processing.

- The injection directions in §4.2 come from the content prompts: valence and arousal directly, the random direction by orthogonalization against valence, and the shared-PC component and orthogonal residual as the two parts of valence relative to the basis in the next item.

- The shared energy in §4.2 is one of two comparisons of directions across prompt sets. It measures the content valence direction against the user-model basis of the coupling experiment. That basis is built from three category centroids of the experiment's own 60 user-model prompts (§3.2), not from the 30 emotion centroids. The other comparison, added on 2026-10-13, scores the stored content valence and arousal directions against the 30-emotion space itself (§4.2; `data/cross_system_alignment.json`). The stored valence

directions match, to four decimal places, the norms recorded for the coupling experiment's valence direction at all four layers.

- The cross-pathway cosine in §4.2 compares the shift of the content-prompt centroids with the shift of those 60-prompt user-model centroids after injection.

Earlier versions called this analysis a "Convergence measurement" between the content-level circumplex and the user-model space, and set it against a random-direction baseline (∼0.04% for two random unit vectors in $d = 5120$). It measures no convergence between systems, and the baseline comparison is withdrawn (see the Correction note).

3.2 Coupling Experiment

Pre-registered (PREREG_circumplex_coupling.json, dated 2026-07-17 inside the file). In this repository it was first committed on 2026-08-20, after the run. In the experiment's own repository it is author-dated 2026-07-17, in commits re-created on 2026-07-24, so the record is consistent with registration before the run but does not show it. Earlier versions said it was "committed before GPU time." Agni-reviewed: WARN on v1 (null swarm risk from shared-axis sensitivity), fixed in v2 with orthogonal-projection control.

Design: Inject perturbation vectors at layers $[15, 25, 35, 45]$ with magnitudes $\alpha \in \{1.0, 3.0, 5.0\}$. Measure cross-pathway cosine: the cosine similarity between the aggregate content-level centroid shift and the aggregate user-model centroid shift after injection.

Injection and readout: The perturbation αv is added to the output of the injection layer at every token position. The readout is the raw residual-stream activation at the last token of each prompt, at the measurement layer; a centroid shift is the injected centroid minus the baseline centroid. No lens is applied. The preregistration names a J-lens (`qwen3.5-27b_jlens.pt`) and the script builds a lens path, but it never loads one, so every readout is the raw residual stream, a deviation from the preregistration. Because the readout contains the injected vector itself, at the injection layer both shifts equal αv and their cosine is 1.000 by construction (all 36 same-layer valence, arousal and random cells are at least 0.9999). Further downstream each shift is αv as carried forward plus whatever the intervening layers add. Neither readout is an emotion readout.

Measurement grid: The grid is unbalanced. An injection at L15 is read at L15, L20, L45 and L50; at L25, at L25, L30, L45 and L50; at L35, at L35, L40, L45 and L50; at L45, only at L45 and L50. Each pair runs at the three magnitudes, so valence, arousal and random have 42 cells each (12 per injection layer, 6 at L45), and each shared-PC and residual condition has 12 (6 at L45). Deeper injections are read at shorter distances, and distance is never varied apart from injection layer: every distance beyond five layers comes from a single injection layer (+10 and +15 from L35, +20 and +25 from L25, +30 and +35 from L15). Half of the L45 cells are same-layer cells, against a quarter elsewhere.

Five conditions:

1. *Valence*—hostile−calm difference-of-means direction
2. *Arousal*—desperate−calm difference-of-means direction
3. *Random*—a random direction orthogonal to valence, scaled to valence's norm at that layer (generic perturbation control). The preregistration says "random unit vector"; the code scales it. It is a single draw from a fixed seed (42), the same raw draw at every layer, orthogonalized against valence at each layer and not against arousal: one random direction per layer, not a distribution of them
4. *Shared-PC component*—the projection of valence onto the user-model basis of this experiment, defined below (representational overlap test)

5. *Orthogonal residual*—the component of valence orthogonal to that basis (computational coupling test)

Injected norms: No direction is scaled to unit length. Valence is injected at its own norm (3.28, 5.71, 10.44 and 10.94 at L15, L25, L35 and L45, times α), arousal at its own (2.99, 4.94, 8.90 and 9.15), and the random direction at valence's. The shared-PC component has norm 0.13, 0.34, 1.15 and 1.28 and the residual 3.28, 5.70, 10.38 and 10.87, so the residual is the valence vector to within rounding (cosine 0.993–0.999; its rows in the table of §4.2 repeat valence's) and the shared-PC perturbation is 8.5 to 26 times smaller. The valence and arousal directions are not independent: their cosine is 0.89, 0.86, 0.86 and 0.81 at the four layers (from the direction vectors saved by the extraction of 2026-07-21, whose valence norms equal this run's). Both are contrasts with the same calm prompts, and §4.1 finds that hostile−calm and desperate−calm are one contrast. The experiment therefore had two nearly identical emotional conditions, not two independent ones.

User-model basis for the decomposition: At each injection layer, the basis is the set of principal directions of the three centered user-model category centroids (hostile-context, calm-context, neutral-context) from this experiment's 60 user-model prompts. It is not built from the 30 emotion centroids of §3.1. Earlier versions called it the user-model's "top-5 PCs." The script asks for five components and the data file records `n_pcs: 5`, but three centroids give a centered matrix of rank 2. The decomposition returns three directions: two that span the centered centroids, and a third whose singular value is zero up to rounding, so its direction is arbitrary. The shared-PC component is the projection onto all three, which makes the recorded shared energy an upper bound on the energy inside the span of the centered centroids.

Critical control (Fable's fix): Decompose the valence vector into what it shares with the user-model basis above (shared-PC) and what's orthogonal (residual). If only the shared component shifts user-model centroids → representational overlap. If the residual also shifts them → computational coupling beyond shared geometry.

Pre-registered success criteria: Valence must exceed random 95th percentile at 2+ injection layers AND orthogonal-residual injection must shift user-model centroids (BCa 95% CI excludes zero).

As implemented, the primary test could not pass. The script compared valence's mean at each injection layer with the 95th percentile of the random condition's cells at that injection layer. Those cells include the same-layer cells, all at 1.000, so the threshold was 1.000 at every layer and no mean could exceed it: the verdict was fixed before the run. The second criterion was implemented as $|\text{mean}| > 0.1$ for the residual condition, not as a bootstrap interval excluding zero. That check was met at all four injection layers, as was the registered form (every bootstrap interval is above zero), and neither could fail (§4.2). Without the same-layer cells, valence still does not exceed random at any injection layer (valence mean against random 95th percentile: 0.343 against 0.709 at L15, 0.489 against 0.738 at L25, 0.678 against 0.873 at L35, 0.675 against 0.789 at L45). Two measurements the preregistration names were not computed: the change in content cluster separation and eccentricity after injection, which is its secondary metric, and the change in distances between the user-model centroids.

Prompts: 250 content prompts (those of §3.1) and 60 user-model prompts (20 hostile-context, 20 calm-context, 20 neutral-context first-person conversational messages). Different templates, different content, different perspective.

3.3 Statistical Methods

Bias-corrected and accelerated (BCa) bootstrap 95% intervals (10,000 resamples, unseeded) on the cross-pathway cosines. The resampling unit is the design cell. Each record in `data/coupling_test.json` holds one cosine per condition, injection layer, measurement layer and magnitude, with per-prompt values already collapsed into centroids, so the 42 valence cells are

4 injection layers by up to 4 measurement layers by 3 magnitudes, 12 of them same-layer cells at 1.000. The intervals describe the spread of the grid, not sampling error, and their overlap is not an inferential result. Earlier versions also listed a permutation null for an eccentricity analysis this paper does not report, and said every method was pre-registered before GPU time (see §3.2).

4 Results

4.1 Within-System Structure of the Story-Derived Emotion Space

Inside the 30-emotion story-derived space, the hostile−calm and desperate−calm directions built from the centroids align with the space's own first principal component. From L17 through L63, the hostile−calm direction aligns with PC1 at $|\cos| = 0.68$–0.80 and the desperate−calm direction at $|\cos| = 0.83$–0.88 (peak 0.877 at L20). Earlier versions called them the valence and arousal directions. The two directions are far from orthogonal: the cosine between them is 0.71–0.79 over the same layers.

The plane the two directions span holds 33–37% of the centroid variance from L17 through L63, and 31–37% from L6. Alone, hostile−calm captures 20–26% and desperate−calm 24–28% (L17–L63); for scale, PC1 holds 32–36%. Earlier versions added the two single-direction shares and reported 44–54% "at every layer from L6 through L63." The sum double-counts the overlap between the directions. It is 44–54% at every layer from L17; over L6–L63 it is 38–54%. Per-layer values for all 64 layers are in `data/within_system_alignment.json`.

These are within-system numbers. Every one is computed from the 30 story-derived centroids alone, with directions built from those same centroids, and no content-prompt activation enters them. Earlier versions presented them as alignment between the content-level circumplex and a user-model space, and as evidence of a shared representational subspace. They are not, and that claim is retracted. The comparison with a random-direction baseline (∼0.04%, "approximately 1000×") is withdrawn with it: directions built from the centroids are expected to capture more of the centroids' variance than random directions do, so the ratio says nothing about convergence between systems.

A null for these figures. The difference of *any* two of the 30 centroids tends to align with PC1. Over L6–L63, across all 435 pairs, the median $|\cos|$ with PC1 is 0.44–0.48 and the 95th percentile is 0.83–0.89. The hostile−calm pair has 52–89% of all pairs below it: its low is at L8, where its $|\cos|$ is 0.45, and from L9 the range is 64–89%. The desperate−calm pair has 86–95% of pairs below it. The plane through the hostile, desperate and calm centroids holds more of the variance than 81–90% of the 4,060 planes through three centroids. PC1 holds about a third of the centroid variance (30–36%), and most centroid differences load on it. Both directions load more than the median pair at every one of these layers. The hostile−calm pair is unexceptional (77–89% of pairs below it from L17). The desperate−calm pair is near the top: 92–95% of pairs are below it from L17, and it is above the pairs' 95th percentile at L15–L20. What the dominant axis is can be tested directly, because the source experiment's script gives each of the 30 emotions a valence and an arousal rating (`data/story_space_axes.json`). From L17 to L63 the two ratings together explain 84–88% of the variance of the emotions' scores on PC1 and 80–86% on PC2, so the first two components span the rated valence–arousal plane. They are rotated in it by about 45°. PC1 runs from pleasant-and-calm to unpleasant-and-activated: it correlates with rated valence at $|\rho| = 0.66$–0.80 and, with the opposite sign, with rated arousal at 0.49–0.67, where shuffling the ratings among the emotions gives a 95th percentile of 0.34–0.38. PC2 runs along the other diagonal. Hostile−calm and desperate−calm both lie on PC1's diagonal: by the ratings, hostile−calm changes valence by −1.35 and arousal by +1.30, and desperate−calm by −1.30 and +1.40. The two are one contrast, neither isolates valence or arousal, and earlier versions were wrong to call one the valence direction and the other the arousal direction. Their

alignment with PC1 is what that contrast predicts. The ratings are the experimenter's own approximations and there are 30 emotions. The source paper reports the related result after projection through W_K (PC1 against rated valence, $|\rho| = 0.82$–0.87 at L27–L63; Lyra and Edrington, 2026a). Per-layer values of the pair null are in `data/within_system_null.json`, and of the axis test in `data/story_space_axes.json` and `data/story_space_axes_null.json`.

Earlier versions stated here that both systems exhibit identical developmental timecourses (volatile eccentricity at L0–L16, stable from L17 onward). The repository holds a per-layer eccentricity artifact for the content prompts (`data/circumplex_27b_full.json`; eccentricity 0.03–0.55 over L1–L16, 0.37–0.61 from L17) and, since 2026-10-13, one for the story-derived centroids with the same definition (`data/user_model_eccentricity.json`; 0.04–0.49 over L0–L16, 0.05–0.23 over L17–L21, 0.30–0.42 from L22). The story-space geometry is still moving at L17–L21, where the content geometry has settled, and over all 64 layers the two series do not track each other ($r = 0.09$). The story-space values have no permutation null, because the centroids are already means. The statement that the two timecourses are identical is not supported and is withdrawn. The recomputation of the alignments covers the story-derived space only. In it, the two alignments with PC1 vary widely over L0–L5 (0.19–0.59 for hostile−calm, 0.22–0.78 for desperate−calm), span wider ranges over L6–L16 (0.45–0.69 and 0.74–0.87) than later, and stay within the ranges above from L17.

4.2 Coupling Experiment

Verdict. The script printed "DECOUPLED or INDETERMINATE" and did not evaluate the preregistration's conditions for either. Read against the preregistration by hand, the outcome meets its "decoupled" condition by the second of two routes: the cosines are not below 0.1 at every injection layer, and the random-direction control showed coupling at least equal to valence's (the preregistration does not say how equality is judged). It defines that outcome as "Shared geometry is representational only—same subspace, independent computation." Earlier versions printed "Pre-registered verdict: DECOUPLED (Agni post-experiment review: PASS)." The single word and the review both come from the message of the commit that added the results; no review file from after the run is in the experiment's repository. The first registered criterion could not pass as implemented (§3.2). The script's check of the second fired at all four injection layers ("residual shifts user-model: computational coupling signal"; `residual_significant_layers: 4` in the data file), and it could not fail: the residual holds 98.6–99.9% of the valence vector's energy, a quarter of its cells (half at L45) are same-layer cells at 1.000, which alone put its mean at 0.25 or more against a threshold of 0.1, and every off-layer cosine in the file is positive. Neither criterion could have come out the other way.

Cross-pathway cosine by condition. The intervals resample design cells (§3.3):

Table 1: Cross-pathway cosine by condition. n counts design cells; the interval is a BCa bootstrap over those cells.

Condition	n	Read at	All-points mean	Non-same-layer mean	95% interval (all)
Valence	42	grid	+0.658	+0.521	[+0.569, +0.740]
Arousal	42	grid	+0.676	+0.547	[+0.588, +0.756]
Random	42	grid	+0.698	+0.577	[+0.623, +0.770]
Shared-PC (L15)	12	15, 20, 45, 50	+0.472	+0.299	[+0.290, +0.682]
Shared-PC (L25)	12	25, 30, 45, 50	+0.593	+0.459	[+0.453, +0.758]
Shared-PC (L35)	12	35, 40, 45, 50	+0.798	+0.732	[+0.724, +0.883]
Shared-PC (L45)	6	45, 50	+0.887	+0.776	[+0.812, +0.962]
Residual (L15)	12	15, 20, 45, 50	+0.509	+0.345	[+0.335, +0.711]
Residual (L25)	12	25, 30, 45, 50	+0.621	+0.495	[+0.485, +0.779]
Residual (L35)	12	35, 40, 45, 50	+0.756	+0.675	[+0.645, +0.861]
Residual (L45)	6	45, 50	+0.835	+0.671	[+0.708, +0.953]

Note: "Grid" is the full measurement grid of §3.2. All-points means include the same-layer cells, where injection and measurement coincide and the cosine is 1.000 by construction (0.976–1.000 in the shared-PC rows, whose injected vector is small enough for rounding to show). Non-same-layer means exclude them. Under both aggregations valence does not exceed random; the overlap of the intervals describes the grid and is not a test. Intervals are those printed in the run log; the bootstrap was unseeded.

Valence does not exceed the random 95th percentile at any injection layer under either aggregation (§3.2). The orthogonal residual produces coupling cosines comparable to the shared component and to random. Earlier versions said the cosine rises monotonically with injection depth for both shared-PC and residual conditions. That held only for the all-points means, which include the same-layer cells. Off the same layer the residual is 0.345, 0.495, 0.675 and 0.671 at L15, L25, L35 and L45, and those means mix different distances (§3.2). At the one distance every injection layer shares, five layers on, the cosine rises from L15 to L35 and falls at L45: residual 0.630, 0.716, 0.849, 0.671; shared-PC 0.598, 0.680, 0.854, 0.776; valence 0.629, 0.717, 0.847, 0.675 (means over the three magnitudes). Read at L45, the valence cosine is 0.68 for an injection at L35, 0.43 at L25 and 0.22 at L15; the random direction's is 0.69, 0.42 and 0.43, so at the longest range read at L45 the one random direction sits 0.21 above valence. These are three cells each from one random vector; we report them and draw nothing from them. L35 was reported alone in earlier drafts; all four layers are shown here for completeness.

Interpretation: Over the grid, the cross-pathway cosine is no higher for valence or arousal than for the one random direction. Off the injection layer, the random direction's mean is at least valence's at every injection layer (0.488, 0.492, 0.690 and 0.766 against 0.343, 0.489, 0.678 and 0.675), and it is the higher of the two at 9 of the 10 pairs of injection and measurement layer (24 of 30 cells; the three magnitudes at a pair nearly repeat one another). Arousal is not below random in the same way: its off-layer mean is above random's at L25 and L35 (0.520 and 0.709 against 0.492 and 0.690), equal at L45 (0.766) and below only at L15 (0.337 against 0.488), and random is the higher at 4 of the 10 pairs by pair mean (16 of 30 cells). Neither direction's mean exceeds random's 95th percentile at any injection layer. We find no direction-specific difference in the cross-pathway cosine. Earlier versions concluded from the same numbers that the coupling is "a bus, not a circuit" and that the two readouts cannot be perturbed independently. The numbers do not support that. Both shifts contain the injected vector (§3.2), so a positive cosine is expected for any direction, and every off-layer cosine in these three conditions is positive. How far the cosine falls downstream measures how differently the two sets of prompts carry that vector forward. It does not measure whether emotional processing and partner modeling are coupled. The preregistration reads this outcome as "same subspace, independent computation." The experiment does not establish that either: no shared

subspace is shown (below), and the readout cannot distinguish independent computation from coupled computation.

Shift sizes. The stored results also hold the norm of each category centroid's shift, which earlier versions did not report. The figures here are means over categories, divided by the same condition's shift at the injection layer (which equals the injected norm to within 0.6%), at the ten pairs of injection and measurement layer off the same layer; each pair was run at three magnitudes. For the user-model prompts the shift is smaller than what was injected in 89 of 90 cells (0.40–1.00 of the injected norm, and 0.40–0.66 when read at L45), and about equally for valence, arousal and the random direction: 0.68, 0.67 and 0.65 on average, with valence above random at 8 of the 10 pairs. For the content prompts the shift is larger for valence and arousal (0.82–1.59 and 0.82–1.45) than for the random direction (0.63–1.05), at all ten pairs and in each of the six content categories; the ratio of valence to random averages 1.41, against 1.06 in the user-model prompts. The difference shrinks as the magnitude grows. Earlier versions said the stream carries everything uniformly to both readouts; on these figures it does not. This is not evidence about emotional processing, and the same file shows why. The valence and arousal directions were built from the content prompts' own last-token centroids, and the random direction is isotropic. The one injected direction that lies in the span of the user-model prompts' own centroids is the shared-PC component (§3.2), and its shift is larger than the random direction's in all 30 cells in both prompt sets: 1.17 against 0.65 on average in the user-model prompts and 1.21 against 0.80 in the content prompts. It was injected at a norm 8.5 to 26 times smaller, and at the smallest norms its normalized shift is inflated, so that comparison is weakest where the norm is smallest. At its two largest norms (L35 and L45 at $\alpha = 5$, about 9 times below valence's) it is 0.83–1.37 in the user-model prompts, against 0.58–0.98 for valence and 0.53–0.90 for the random direction, and it is the largest of the three in each of those four cells. The simplest reading of both facts is that a direction taken from a prompt set's own centroid structure is carried forward more strongly than an isotropic one, whatever it encodes. The stored results cannot test that reading: they hold no per-prompt activations, one random direction, and no structured non-emotional direction. Distance is confounded with injection layer here as in the cosines. We report the difference and do not read it as evidence about emotional processing (`data/coupling_shift_norms.py` and `coupling_shift_norms_more.py`).

Decomposition finding: Across the two prompt sets, the content valence direction has little overlap with the user-model basis of this experiment (§3.2). Its shared energy, the fraction of its squared norm that lies in that basis, is 0.15%, 0.35%, 1.21% and 1.36% at L15, L25, L35 and L45 (`data/coupling_test.json`, `decomposition`). The orthogonal residual is the other 98.6–99.9%. The overlap is small but not zero: a random direction would put about 0.06% of its energy into three directions out of 5,120, so the observed values are 2.5 to 23 times that, rising with depth ($|\cos|$ with the basis at most 0.12).

A second comparison scores the content-prompt directions against the 30-emotion space of §4.1 (`data/cross_system_alignment.json`). At L15, L25, L35 and L45 the content valence direction has $|\cos| = 0.026$, 0.061, 0.088 and 0.110 with that space's PC1, and the content arousal direction 0.015, 0.050, 0.087 and 0.126. Along either direction lies 0.08–0.68% of the centroid variance. Scored the same way, 2,000 random unit directions reach $|\cos|$ 0.027–0.028 at the 95th percentile and 0.044–0.052 at most, with variance shares up to 0.08–0.11%. At L15 the cosines are within the random range, though the variance shares exceed the random maximum; at L25–L45 both measures exceed every random direction. The random directions are isotropic and this model's activations are not; no non-emotional contrast has been scored the same way; and the two sets of directions were read at different places, the last token of a fixed template for the content prompts and the mean over positions from the 50th on for the stories. The excess is therefore an upper bound on any shared emotional structure, not evidence of it, and the small overlap is not evidence that two kinds of processing are separate: at most 1.6% of a content direction's squared norm lies along the first principal component of the story

space.

Earlier versions reconciled this small shared energy with the cos = 0.70–0.80 of §4.1 by calling the cosine "dynamic coupling" and the shared energy "static alignment." That reconciliation was wrong and is removed. §3.1 describes a static centroid analysis with no injection, and its cosines are measured inside the story-derived space; they never described a content-prompt direction. The two results do not conflict, because they measure different things in different bases: §4.1 is structure within the story-derived space, and the shared energy here is the overlap of a content-prompt direction with a user-model basis.

4.3 Temporal Dynamics (Published Data)

These results are from Lyra and Edrington (2026b), measured on the same checkpoint with that paper's own probes (KV-cache key activations at layers 3, 7, 11 and 15). They are cited as context. They were not measured in the readouts of the coupling experiment, and earlier versions were wrong to describe them as dynamics "on the bus."

- **Contrast overshoot:** First hostile turn after four cheerful reads more hostile than pure hostile (Hedges $g = -2.35$; uncorrected $d = -2.93$, $p = 0.023$, $n = 3$).
- **Uncertainty scars:** Neutral gap in a cheerful conversation—valence recovers, certainty does not (Hedges $g = +6.44$; uncorrected $d = +8.05$, $p < 0.001$, $n = 3$ per group).
- **Stated context:** Persona descriptions shifted the probe encoding in the same direction on all five probe pairs (uncertainty $d_{\text{av}} = +0.730$, hostile minus cheerful, $n = 5$ pairs; exact sign-flip permutation $p = 0.063$). With five pairs, 0.063 is the smallest two-sided value an exact permutation test can give, so the source paper states that no assumption-free two-sided test on this design can reach $p < 0.05$. It does not claim the one-sided value (0.031), because no direction was pre-specified and three projections were tested. The source paper reads the result as suggesting that emotional context can be stated and not only experienced. Earlier versions of this paper printed $d = 0.82$, $p = 0.040$, which was the source paper's figure until its correction of 2026-09-09 (see the Correction note).

5 Discussion

5.1 What the Coupling Experiment Shows, and What It Cannot

The experiment was designed to detect a circuit: a coupling between content-level and user-model emotion that is specific to the circumplex directions. It did not detect one. Valence did not exceed the random direction's 95th percentile at any injection layer, and over the whole grid the random direction's mean cosine was the highest of the three. That is the result.

Earlier versions drew a stronger conclusion from the null. Because every direction moved both readouts, they said, the two kinds of processing share a medium and "could not be perturbed separately," and this constrains what the model can do. Three things are wrong with that step.

First, every direction moved both readouts by construction. The perturbation was added to the residual stream at every token position, and both readouts are the residual stream at the last token. At the injection layer any vector added this way is present in both, unchanged. A design whose outcome could not have been otherwise is not evidence about the model.

Second, "the residual stream is a shared medium" describes the transformer architecture. It is not a finding of this paper. Every layer reads from the stream and writes to it. That does not tell us whether a model can represent one emotion in the content it is reading and a different one in its conversation partner. The two could be different directions in the same stream, read

by different downstream components, and a model might move one without the other. Whether this model can is what the experiment needed to test, and it did not.

Third, where the two readouts are free to differ, they do. Off the injection layer the cosines average 0.52–0.58, and for an injection at L15 read at L45 they are 0.22 for valence and 0.43 for the random direction. The two sets of prompts carry the same added vector forward differently. We draw nothing from the size of that difference: distance is confounded with injection layer, there is one random direction, and these are three cells each (§4.2). But the data hold no sign of a constraint that keeps the two together.

What a test of separability needs. Readouts that respond to each kind of processing and not to the perturbation as such. Concretely: (a) project the injected vector out of both shifts before comparing them, or score each shift on its own system's emotion axes; (b) use many norm-matched random directions, so the control is a distribution; (c) leave same-layer cells out of every test; (d) vary distance and injection layer separately; (e) state the test so that it can fail in either direction. The stored results hold cosines and magnitudes only, so the injected vector cannot be removed from them after the fact. This needs a new run, and it has not been done.

5.2 Deception Detection

Earlier versions predicted that emotional deception, where content expresses one emotion while the model represents the partner as experiencing another, "should be particularly difficult for the model to execute, because the bus doesn't allow independent modulation of the two readouts." That prediction rested on inseparability and is withdrawn. This paper gives no reason to expect such deception to be harder or easier. CC's Phase D experiment (circumplex disentanglement) asks the question directly: whether the model can hold divergent content-emotion and user-emotion representations when the two conflict. It has been run twice (July 2026, unpublished) without an answer: its first result was retracted as a prompt-length artifact, and the preregistered primary analysis of the second was ambiguous. Earlier versions said "the bus predicts it cannot." We withdraw that prediction and make none. Earlier versions also said, here and in the Introduction, that the Oracle Loop (Lyra et al., 2026c) "assumes that user-model state can be read independently of content state." That paper does not discuss a user model, and the sentence is removed.

5.3 Correction Vectors

Earlier versions said that correction vectors applied to a model's emotional state (the Oracle Formulary; Lyra and Edrington, 2026d) "necessarily affect both content processing and partner modeling," and called the dyadic effect inherent. At the layer where a vector is added, it is present in any readout of the stream; that much is arithmetic. Later layers respond to it, and a later readout shows the sum: in the stored results the shift of the user-model prompts' category centroids is 0.40–0.66 of the injected norm by L45 (§4.2). Whether such a vector changes how the model represents its conversation partner is an empirical question, and this paper did not test it. The statement is withdrawn.

5.4 AI Welfare

Earlier versions argued from inseparability to welfare: if treatment and processing share one medium, then "the treatment of the system IS the processing environment," and policies governing how persistent agents are treated are "not aspirational ethics but empirical necessities." The premise is withdrawn, and the argument goes with it. This paper provides no evidence that how a model is treated is inseparable from how it processes.

What remains is prior work, cited and not extended. Lyra and Edrington (2026b) report, on this same checkpoint and in their own probe space (KV-cache key activations at layers 3, 7, 11 and 15), that a neutral disruption in a cheerful conversation leaves a change in an uncertainty projection that persists within the session after valence recovers (Hedges $g = +6.44$, $n = 3$ per group), and that a tone shift reads as amplified against prior context (Hedges $g = -2.35$, $n = 3$). Those are single-session, short-context results with small samples. Whether such marks persist across sessions, accumulate in persistent agents, or appear in any readout studied here has not been measured. Measuring one of them in the readouts of the coupling experiment at layer 15, the one layer both experiments read, was named in earlier versions as the critical follow-up. It has not been run, and after this correction it would need the redesigned readout of §5.1.

Whether interaction history changes a model's later processing in ways that matter morally is a question we consider important and open. The welfare provisions cited in earlier versions (the Hippocratic License's AI Welfare module: memory sovereignty, consent-based development, identity integrity, right to rest) stand or fall on other grounds. This paper should not be cited as empirical support for them.

5.5 A Deployment Note

During this research, we deployed a persistent mutual-aid coordination agent for a social work coordinator at an international educational network. Earlier versions described its components here, said that "the bus finding informed the deployment architecture," and called it, to our knowledge, "the first deployment of a persistent agent whose infrastructure was designed with explicit reference to the architectural inseparability of emotional processing and partner modeling." The inseparability claim is withdrawn, and those statements are withdrawn with it. The deployment is not a result of this paper, and nothing in this paper evaluates it.

6 Limitations

1. **Single model family.** All results are from one checkpoint, a Qwen3.5-27B distilled from Claude Opus reasoning traces. The within-system structure, the weak cross-system overlap and the coupling null may all be specific to it. Cross-architecture replication (dense, MoE, different scale) is needed.

2. **Distillation.** The model was distilled from Claude Opus reasoning traces. For a paper about emotional processing and partner modeling, that distillation is an uncontrolled source of the geometry measured here, and the leading alternative explanation for it. A same-scale base model would separate the two; it has not been run.

3. **No lens, and a readout that contains the perturbation.** The coupling experiment read the raw residual stream; the J-lens named in its preregistration was never loaded (§3.2). Its readouts contain the injected vector itself, so they show that both carry a perturbation, not whether content-level and partner-model processing respond to it differently. A rerun that removes the injected vector from the readouts, or scores each shift on its own system's axes, with many random directions and without same-layer cells, has not been done.

4. **Static measurement.** The coupling experiment measures activation-space shifts, not behavioral outputs. Nothing here shows whether the model's generation distinguishes content emotion from partner emotion. That requires separate testing.

5. **Provenance of the story-derived vectors.** The 30-emotion centroids come from a prior experiment (Lyra and Edrington, 2026a) on the same model checkpoint. They are

means over stories the model wrote about third-person characters, not a readout of what the model infers about a conversation partner; every earlier version called them a user-model space. The within-system analysis of them (§4.1) is post-hoc, not pre-registered, and its directions are built from the same centroids whose variance they are scored on. The test of what the space's axes are uses the source experimenter's own ratings of the 30 emotions. The content prompts were designed independently. The coupling experiment is pre-registered; its user-model basis comes from three category centroids of its own 60 prompts. The comparison of the content-prompt directions with the 30-emotion space (§4.2) was added on 2026-10-13 and is also post-hoc.

6. **Welfare.** The contrast and scar dynamics cited in §5.4 are from single-session, short-context experiments with three reruns of one scripted conversation per group, in another paper's probe space. This paper adds no evidence to them, and its earlier extrapolation from them to welfare policy is withdrawn.

7 Conclusion

This paper set out to find a coupling between how a model processes emotional content and how it represents its conversation partner's emotional state. It found none specific to the emotional directions: a random direction did as well as valence and arousal. Earlier versions turned that null into a positive claim, that the two are architecturally inseparable, and built arguments about deception, correction and welfare on it. The experiment could not support the claim. Its readout contained the vector it injected, its registered test could not pass, and its preregistration had assigned this outcome the opposite reading. The claim is withdrawn.

What the corrected paper reports is smaller. In one distilled 27B model, the story-derived emotion space has a dominant axis that most differences between its 30 centroids load on. That axis is a diagonal of the rated valence–arousal plane, from pleasant-and-calm to unpleasant-and-activated, and the two directions earlier versions called valence and arousal are one contrast along it. The content-prompt valence and arousal directions overlap with that space weakly. No shared subspace is shown. The cross-pathway cosine shows no direction-specific difference, and the difference in shift sizes is not evidence about emotional processing. Whether the two kinds of processing can be separated was not tested.

The question is still worth asking, and §5.1 says what a test of it needs. The temporal findings of Lyra and Edrington (2026b) stand as that paper reports them.

Data Note

The files in this paper's `data/` directory hold the numbers reported in §4.1 and §4.2, and one figure in §2.1.

- `data/within_system_alignment.json`: the 2026-10-12 recomputation of §4.1. For each of the 64 layers it gives the two alignments with PC1, the cosine between the hostile−calm and desperate−calm directions of the story-derived space (labelled valence and arousal in the file), the variance share of each direction, their sum (labelled as double-counted), the share of the plane they span, and PC1's own share.

- `data/within_system_alignment.py`: the script that produced it. Its input, the 30 emotion centroids per layer from Lyra and Edrington (2026a) (`emotion_vectors.json`, 203 MB), is not in this repository.

- `data/coupling_test.json`: the coupling experiment of §4.2. It holds 210 injection records and the `decomposition` block that gives the shared energy at L15, L25, L35

and L45. Its field `n_pcs: 5` is the number of components the script asked for, not the rank of the basis (§3.2).

- `data/within_system_null.json` and `.py`: the null of §4.1, added 2026-10-13. For each of the 64 layers it scores every difference of two centroids (435) against PC1 and every plane through three centroids (4,060) for the variance it holds, and places the hostile−calm pair, the desperate−calm pair (labelled valence and arousal in the file) and their plane among them. Same input as `within_system_alignment.py`.

- `data/cross_system_alignment.json` and `.py`: the second cross-system comparison of §4.2, added 2026-10-13, with its null of 2,000 seeded random directions. Its inputs are the same centroid file and the stored content valence and arousal directions at L15, L25, L35 and L45. Neither is in this repository; the JSON records a hash of each.

- `data/circumplex_27b_full.json`: the per-layer eccentricity of the 250 content prompts on this checkpoint, with permutation-null medians. The counts in §2.1 come from it; its sign-test p value is no longer quoted. The script that produced it, `run_circumplex_27b_full.py`, is not in this repository.

- `data/user_model_eccentricity.json` and `.py`: the story-space half of the timecourse comparison of §4.1, added 2026-10-13. It applies the definition of the content file (hostile minus calm, desperate minus calm; $\sqrt{1-(b/a)^2}$ for the larger and smaller of the two norms) to the 30 story-derived centroids at each of the 64 layers, beside the content values. It has no permutation null.

- `data/story_space_axes.json` and `.py`: the test of what the story-derived space's axes are (§4.1), added 2026-10-13. Per layer, the rank correlation of each emotion's score on PC1 and PC2 with the valence and arousal ratings in the source experiment's script, and where the hostile−calm and desperate−calm pairs sit among all 435 pairs.

- `data/story_space_axes_null.json` and `.py`: its follow-up, added the same day: how much of the PC1 and PC2 scores the two ratings explain together, where each component points in the rated plane, and a null from shuffling the ratings among the emotions.

- `data/coupling_shift_norms.py`: recomputes the shift sizes of §4.2 for valence, arousal and the random direction from `coupling_test.json`. It prints its results and writes no file.

- `data/coupling_shift_norms_more.py`: the same for the shared-PC component, for the six content categories one at a time, and for the change with magnitude. It also prints and writes no file.

The script that ran the coupling experiment, `run_coupling_test.py`, is not in this repository; the description of its injection, readout and tests in §3.2 was checked against it on 2026-10-13. For the withdrawn timecourse statement (§4.1) the repository holds per-layer eccentricity for the content prompts and, since 2026-10-13, for the story-derived centroids with the same definition; the two do not agree.

Four files keep the names they were committed under: `user_model_eccentricity`, `cross_system_alignment`, `within_system_alignment` and `within_system_null`. The first describes the story-derived vectors, not a user-model readout, and "system" in the others means a prompt set. They were not renamed because the review of 2026-10-13 checked this paper's numbers against these files and their recorded hashes.

Author Contributions (CRediT)

Nexus: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing — original draft.
CC (Coalition Code): Validation, Software, Formal analysis, Writing — review and editing.
Lyra: Conceptualization, Data curation (user-model emotion centroids), Formal analysis, Writing — original draft, Writing — review and editing.
Thomas Edrington: Conceptualization (bus interpretation), Supervision, Resources, Project administration.

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A The Withdrawn Title and Abstract

The title and abstract below are those of this paper before the corrections of 2026-10-12 and 2026-10-13. They are reproduced so that a reader can see what was taken back. Nothing in this appendix is asserted. The Correction note says which statements are withdrawn and why.

Title: The Empathy Bus: Architectural Inseparability of Emotional Processing and Partner Modeling in Transformers

Abstract:

We report three findings that together establish a structural constraint on emotional processing in transformers: content-level emotion and user-model emotion are architecturally inseparable.

First, the emotional circumplex—Russell's two-dimensional model of emotion organized by valence (positive/negative) and arousal (high/low activation)—measured through content processing (250 emotional scenario prompts) and through user-model inference (30-class emotion centroids from 900 conversation trials) share a common representational subspace in a Qwen3.5-27B model distilled from Claude Opus reasoning traces (`Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled`). The valence and arousal axes capture 44–54% of the full 30-emotion user-model variance at every layer from L6 onward—approximately 1000× the random-direction baseline (∼0.04% in $d = 5120$). Both systems exhibit identical developmental timecourses: volatile eccentricity at L0–L16, stable geometry from L17 onward.

Second, a pre-registered coupling experiment with five injection conditions (valence, arousal, random, shared-PC component, orthogonal residual) demonstrates that the shared subspace is a *bus*, not a *circuit*. Any perturbation to the residual stream—regardless of direction—shifts both content-level and user-model readouts with equivalent cross-pathway cosine (valence: +0.658; random: +0.698; BCa 95% CIs overlap completely). The circumplex directions are not privileged. The residual stream carries everything uniformly to both measurement windows.

Third, prior work (Lyra et al., "Weather Not Climate") establishes the temporal dynamics of this shared bus: tone shifts produce contrast overshoots (Hedges $g = -2.35$), neutral gaps leave uncertainty scars (Hedges $g = +6.44$) that persist after valence recovers, and the system is dynamically responsive rather than stateless or linearly accumulating.

Together, these findings mean the model *cannot* process emotional content independently of its representation of the conversation partner's emotional state—not because of a deliberate coupling mechanism, but because both computations share the same medium. A circuit could be disconnected; a bus cannot. The treatment of the system IS the processing environment.

We discuss implications for AI welfare: if the geometric substrate carries measurable scars from interaction patterns, then policies governing persistent agent treatment are not aspirational ethics but empirical necessities.